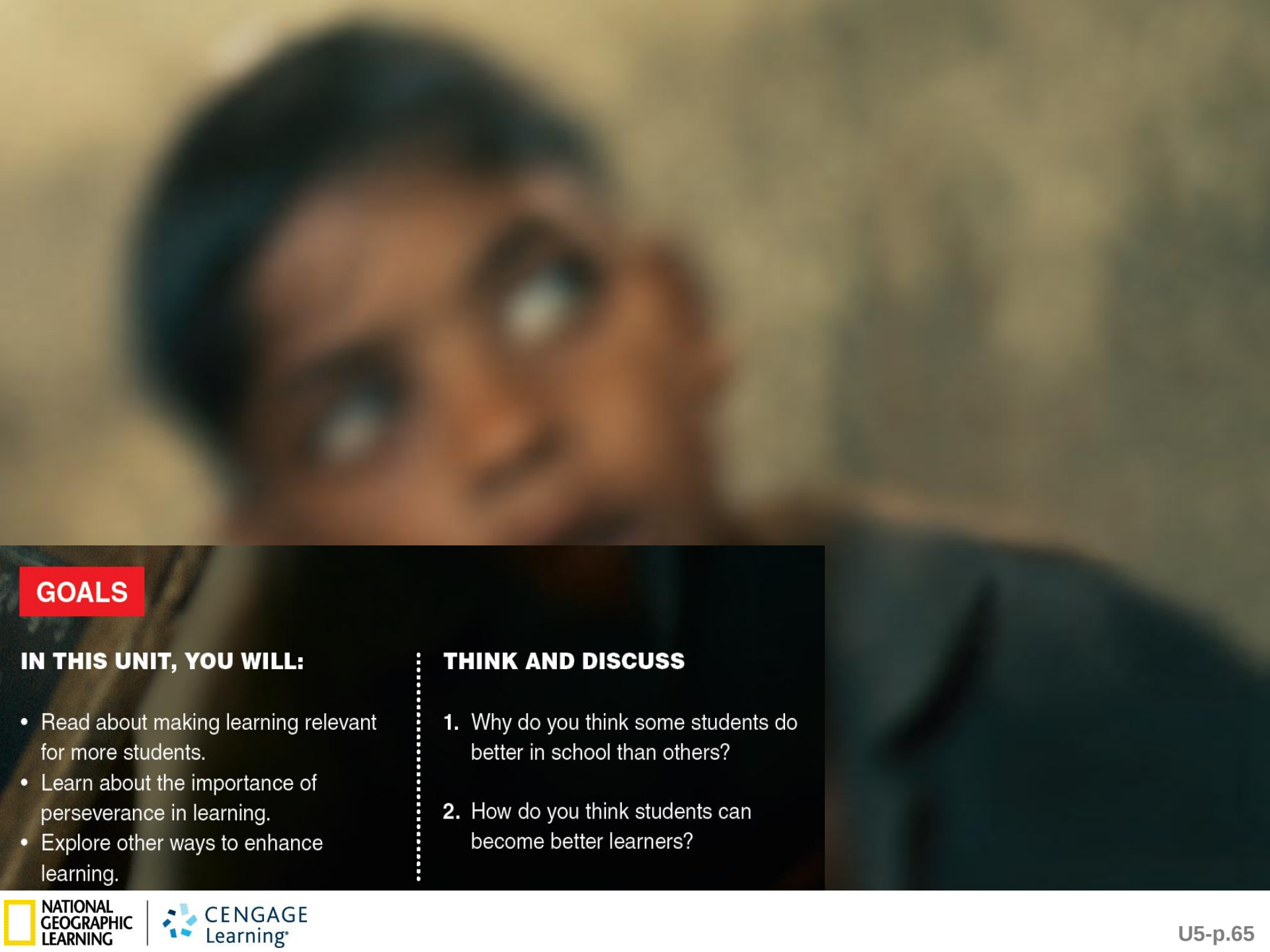


A boy pays close attention in a classroom in Odisha State, India, in 1970.

# LESSONS IN LEARNING



## GOALS

### IN THIS UNIT, YOU WILL:

- Read about making learning relevant for more students.
- Learn about the importance of perseverance in learning.
- Explore other ways to enhance learning.

### THINK AND DISCUSS

1. Why do you think some students do better in school than others?
2. How do you think students can become better learners?

# Lesson A

## PRE-READING

- A.** Look at the charts on page 67. Why do you think more students feel disengaged with school as they get older? Discuss with a partner.

*Answers will vary. As students get closer to the age of independence, when they'll find jobs and support themselves, many start to feel less connected to the school environment and the traditional learning process found in most schools.*

**B.** Look at the information at the bottom of page 68 about The Da Vinci Studio School of Science and Technology. Then answer the questions. Discuss your answers with a partner.

1. How is the Da Vinci school different from traditional schools?

The Da Vinci School combines traditional education with practical on-the-job training.

2. Why do you think students at this school don't have to do homework?

It is likely that they don't have homework because they also have jobs.

3. What do you think a "personal coach" does?

A personal coach helps guide and advise an individual to help that person succeed, in this case at meeting their learning goals.

**C.** What makes some schools more appealing to students than other schools? Write some notes below and discuss with a partner.

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Check your ideas as you read the article.



High school students look at water samples on a field trip in Bayou La Branche, Louisiana, U.S.A.

# ENGAGING LEARNERS

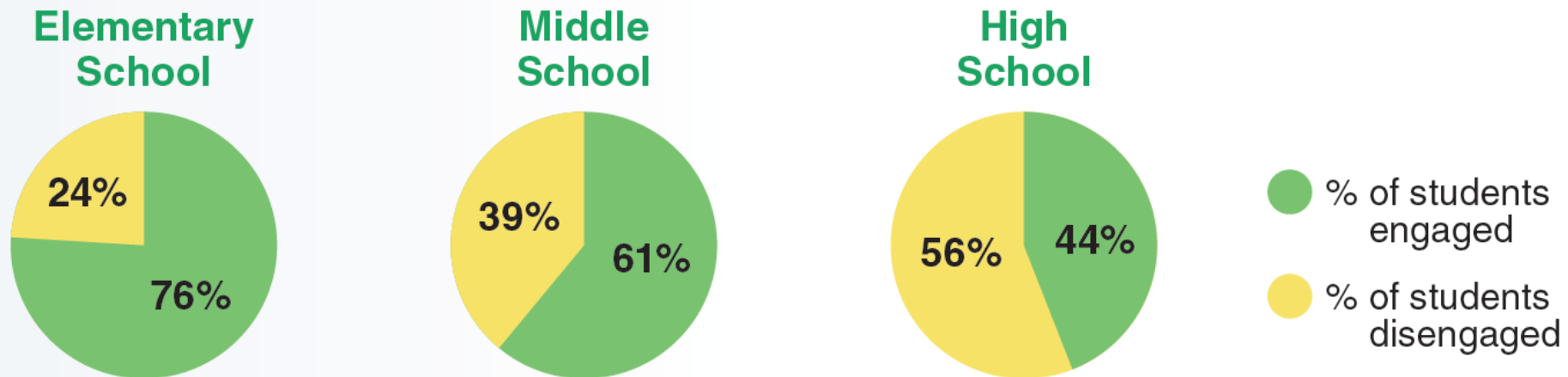


Not all teenagers do well at school, but why is that? There are plenty of theories—poverty, I.Q. family background. But researchers are also discovering that other factors lie at the heart of student success.



A recent Gallup poll in the U.S.A. found that over half of all high school students do not feel **engaged** with their studies. Why? Researchers think some students do not thrive in schools that focus on teacher-centered, test-focused, and academic learning. These students often struggle to see the relationship between this kind of learning and their lives. A report from the U.K. **indicates** that nearly a third of students do not complete their education because they don't think their schooling is **relevant** to the kind of work they will do in the future.

## Students' Engagement Over Time



Source: Gallup, 2013



How can teenage students become more engaged? Geoff Mulgan, social commentator and TED speaker, believes the solution lies in making learning more relevant. He researched how people learn in schools around the world, from Australia to Paraguay. His research suggested that a new model of schooling was needed—a model that connects what students are learning to their own plans for their future. “What kind of school would have teenagers fighting to get in, not fighting to stay out?” wondered Mulgan.

## LEARNING BY DOING

Mulgan helped create the Studio School, a system of publicly-funded schools for 14- to 19-year olds in the U.K. Unlike traditional education, Studio Schools **combine** academic learning with practical training for future jobs. About 80 percent of the curriculum is done not through sitting in classrooms but through real-life, practical projects, including working at local businesses. Mulgan points out that many teenagers “learn best in teams, and they learn best by doing things for real.”



This different **approach** is also enhanced by more personalized instruction and support. Studio Schools are smaller than traditional schools—typically around 300 learners—to ensure that every student feels valued and supported. Each student has a “personal **coach**” who helps him or her develop a plan to meet their learning goals. By having more tangible goals, students become more **determined** to do well at school. Reports have shown **promising** results. Many traditional-school students who were not getting good grades improved significantly after they moved to a Studio School.

Mulgan says Studio Schools are based on a simple idea that turns education on its head. They put teamwork and practical projects at the heart of learning. This simple idea is spreading. From just one school in 2010, Studio Schools had grown to more than 40 by 2014.

---

**curriculum:** *n.* the content and skills taught in a school or college course

**tangible:** *adj.* easy to see or recognize

**apprentice:** *n.* a person who is trained by an expert in a job skill

**fund-raiser:** *n.* a person who collects money for a charity or other organization

# THE DA VINCI STUDIO SCHOOL OF SCIENCE AND TECHNOLOGY



**Dress** Black suit, black shoes, blue “Da Vinci” shirt

**Academic studies** Same as traditional schools, including English, math, science

**Typical work placements** sales apprentice at Ford, assistant at jewelry store, fund-raiser at local charity store

## **Time spent at work placement**

Under 16: one day every two weeks;  
Over 16: two days a week, paid

**Homework** None; all work done in school hours

**Other** Personal coach available all year



**Location** Stevenage, England

**Size** 300 students

**Ages** 14–19

**Hours** 9:00–5:00



A teacher writes on the board in a classroom in Aguas Calientes, Peru.

# Developing Reading Skills

## GETTING THE MAIN IDEAS

Use the information from the passage on pages 66–68 to answer each question.

1. How does the writer describe the approach to learning in traditional schools?  
*teacher-centered, test-focused, academic learning*
2. What is the main reason some students aren't engaged in traditional schools?
  - a. They don't do well at tests.
  - ☒ b. They don't see the connection to their future lives.
  - c. They like doing projects.

3. Why were Studio Schools created?

- a. Too many teenagers were fighting at traditional schools.
- ☒ b. Traditional schools weren't meeting the needs of all students.
- c. There were not enough traditional schools for all students.

4. How might Geoff Mulgan describe Studio Schools? Complete the sentence.

Studio Schools combine academic learning  
with practical training.

# SUPPORTING IDEAS WITH EVIDENCE

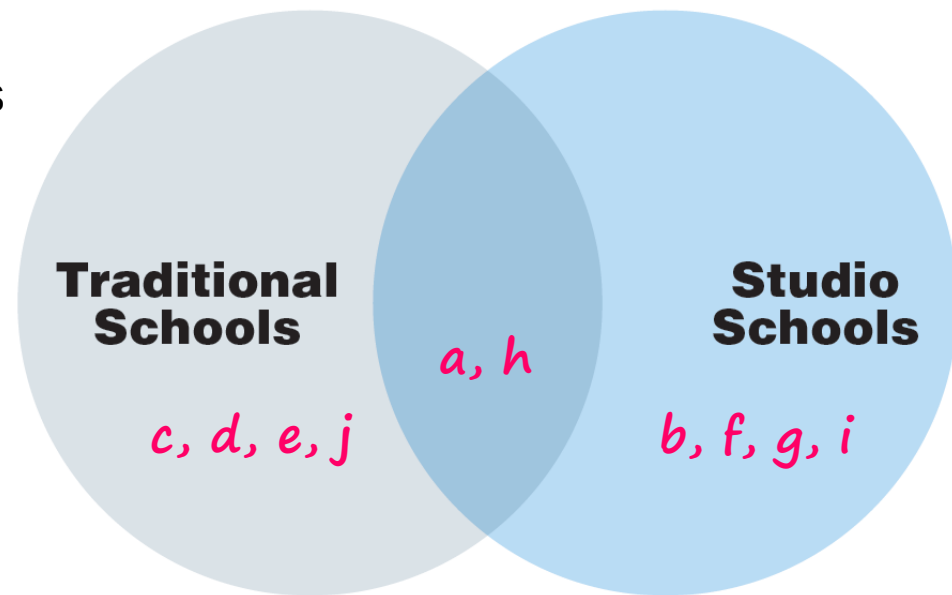
Scan the passage and infographics on pages 66–68, and write answers to the questions below.

1. What percentage of U.S. high school students don't feel engaged with their studies, according to a recent poll? 56 percent
2. How many learners do not finish their education in the U.K.? one-third
3. How many students are in a typical Studio School? 300
4. In Studio Schools, how much of the curriculum is covered outside the classroom? 80 percent
5. How many Studio Schools were there in 2014? more than 40

# FINDING SIMILARITIES AND DIFFERENCES

How are Studio Schools the same as and different from traditional schools? Use the information below (a–j) to complete the Venn diagram.

- a. academic learning
- b. emphasis on teamwork
- c. includes elementary school students
- d. focus on taking tests
- e. homework
- f. learning is mostly project-based
- g. personal coaches
- h. publicly-funded
- i. emphasis on work placements
- j. majority of time spent in a classroom



# GETTING MEANING FROM CONTEXT

What do you think Geoff Mulgan means when he says that a Studio School “turns education on its head”? Discuss your ideas with a partner.

*He means that what is normal is turned upside-down and changed completely.*

---

## CRITICAL THINKING

1. **Personalizing.** Would you like to go to a Studio School? Why, or why not?

*Answers will vary.*

---

2. **Reflecting.** Mulgan says that large numbers of teenagers “learn best in teams, and they learn best by doing things for real.” Do you agree with him? Explain why, or why not.

*Answers will vary.*

---

# BUILDING VOCABULARY

**A.** Complete the paragraph with the correct words.

approach      determined      engage      indicated      relevant

Why do some students drop out of school? Why do some feel it is not relevant to their lives? There are plenty of theories, ranging from poor teachers to bad curricula. However, Rita Pierson, a teacher in the United States for more than 40 years, suggested we take a different approach to this question. Her experience indicated that we should focus more on the human connection. In other words, we should look at how teachers engage with their students. Even simple things, she said, like encouraging students when they fail and being determined to understand their problems, can make a big difference in a student's life.

**B. Choose the best option to answer each question.**

- b* 1. Which of these words best describe something that shows signs of future success?  
a. engaging      b. promising      c. surprising
- c* 2. Which of these verbs best describes joining two or more things together?  
a. create      b. determine      c. combine
- a* 3. Who or what is trained to give advice or instruction?  
a. a coach      b. an approach      c. a child

## EXPLORE MORE

Find out more about Studio Schools. Go to TED.com and watch Mulgan's TED Talk called "A short intro to the Studio School." Why did Mulgan choose "Studio School" as a name for this type of school? Share your opinions of Mulgan's talk with your class.

*In Renaissance times, a studio was a place where learning and working were combined.*



# THE KEY TO SUCCESS? GRIT

**ANGELA LEE DUCKWORTH** Psychologist and academic, TED speaker

Angela Lee Duckworth wants to find out why some students succeed where others fail.

In her late 20s, Duckworth left a job as a management consultant to teach high school math in public schools in San Francisco, Philadelphia, and New York. She realized that the smartest kids were not always the ones who got the best grades.

Duckworth did some research and has proposed some new ideas about success in learning. She thinks there is one factor that affects students' engagement in school. She has won numerous honors and awards for this research. In 2013, she received a Genius Grant from the MacArthur Foundation. Her work has also generated a lot of interest among educators.

**In this lesson, you are going to watch segments of Duckworth's TED Talk. Use the information about Duckworth above to answer these questions.**

1. What did Duckworth realize when she taught high school math?

*The smartest kids are not the ones who get the best grades.*

---

2. What did her studies suggest about success in learning?

*There is one factor that affects engagement in school.*

---

3. What was the reaction from other people to Duckworth's ideas? How do you know?

*People are excited by her ideas. Her various honors and grants demonstrate this.*

---

Duckworth's **idea worth spreading** is that the biggest predictor of success for students is not I.Q., but the ability to persevere and have passion for long-term goals.



## PART 1

# JUST I.Q.?

## PREVIEWING

Read this excerpt from Angela Lee Duckworth's talk. What surprises Duckworth about the difference between her best and worst learners? Note your answer in your own words.

“What struck me was that I.Q. was not the only difference between my best and my worst students. Some of my strongest performers did not have stratospheric I.Q. scores. Some of my smartest kids weren't doing so well.”

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“What struck me . . .”: “What I realized . . .”

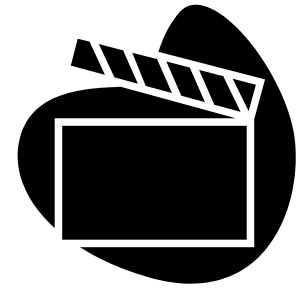
stratospheric: *adj.* very high

I.Q.: *n.* level of someone's intelligence (an abbreviation of intelligence quotient)

---

*Answers will vary somewhat. Duckworth says that the students who do well in school are often not the smartest ones. And some of her worst learners were in contrast very smart individuals.*

---



## GETTING THE MAIN IDEA

Read the excerpt below from Duckworth's talk. Then watch (▶) the first segment of Duckworth's talk. What conclusions did she come to? Check (✓) all that apply.

“After several more years of teaching, I came to the conclusion that what we need in education is a much better understanding of students and learning from a motivational perspective, from a psychological perspective. In education, the one thing we know how to measure best is I.Q., but what if doing well in school and in life depends on much more than your ability to learn quickly and easily?”

---

**motivational perspective:** *n.* understanding what it is that drives people to want to learn

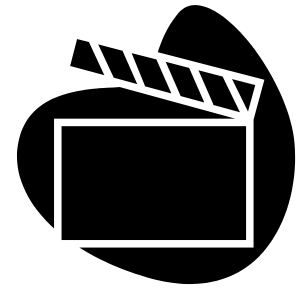
**psychological perspective:** *n.* understanding how social and emotional factors affect learning

- ☒ 1. We need to understand more about students and learning.
- ☐ 2. The school curriculum is too difficult for most students.
- ☒ 3. It's important to look at what makes students want to learn.
- ☐ 4. Schools need to measure students' I.Q. more carefully.
- ☐ 5. Students should be trained to learn quickly and easily.
- ☒ 6. Schools may be putting too much focus on measuring I.Q.

## CRITICAL THINKING

**Predicting.** What other factors do you think Duckworth found that increase success in learning?

*Answers will vary. She reveals in the second part of her talk that grit is the major factor affecting success.*



## PART 2

## TRUE GRIT

## UNDERSTANDING THE OVERALL MESSAGE

Read the questions. Circle the correct answers as you watch (▶) the next segment of Duckworth's talk.

1. Which of these best describes what Duckworth means by *grit*?
  - a. A very challenging situation
  - b. Having good health
  - ☒ c. Hard work and determination
2. Which statement best matches Duckworth's overall message?
  - a. Grit is a more significant predictor of success in learning than I.Q.
  - b. No one really knows why some students are more successful than others.
  - ☒ c. In order to be successful learners, we must sometimes fail and start over.

## UNDERSTANDING TERMS

**Read the excerpt from Duckworth's talk. Then answer the questions below.**

“So far, the best idea I’ve heard about building grit in kids is something called “growth mindset.” This is an idea developed at Stanford University by Carol Dweck, and it is the belief that the ability to learn is not fixed, that it can change with your effort. Dr. Dweck has shown that when kids read and learn about the brain and how it changes and grows in response to challenge, they’re much more likely to persevere when they fail, because they don’t believe that failure is a permanent condition. So growth mindset is a great idea for building grit.”

---

**persevere:** *v.* to continue trying to achieve something even though it is difficult

**permanent:** *adj.* remaining the same and never changing

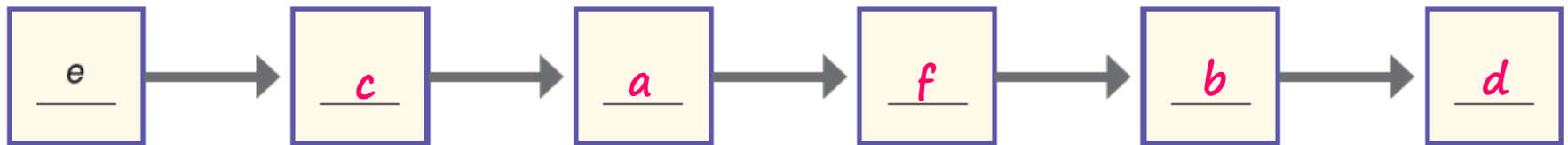
1. What are two effects of teaching children about “growth mindset”?
  - a. They see that effort can change a person’s ability to learn.
  - b. They understand how the brain works.
  - c. They learn that our ability to learn decreases in later life.
  
2. How does an understanding of “growth mindset” help learners to succeed?

When students realize that the human brain grows in response to challenges, they are more likely to keep trying when a challenge or difficulty arises.

## SUMMARIZING THE TALK

Read the statements (a–f). Put them in the correct order in the chart to summarize Duckworth’s talk.

- a. Then she discovers one key factor, which she calls grit.
- b. However, more work needs to be done to find out more about building grit.
- c. She studies lots of learners to figure out what makes them successful.
- d. “Growth mindset” may help build grit.
- e. Duckworth notices that a high I.Q. doesn’t always mean good grades.
- f. Grit doesn’t come from being naturally talented, it comes from working hard.



## GETTING MEANING FROM CONTEXT

Duckworth says that grit is living life like it's “a marathon, not a sprint.” What do you think she means by this? Discuss your ideas with a partner.

*She means that we must see life as a long journey with ups and downs that we must persevere through instead of expecting quick and easy success.*

## CRITICAL THINKING

1. **Synthesizing.** Duckworth and Mulgan each have ideas about why some students aren't successful in school. How are their ideas similar or different? Explain your answer.

*Both say that traditional learning is not the best for all students. Duckworth focuses on a character trait that she wants to develop more in students, while Mulgan focuses on how schools can engage learners by changing the way teaching and learning happen.*

2. **Reflecting.** Do you think that grit is useful in other areas of life, or does it just apply to students in school? Explain your ideas.

*Answers will vary. However, Duckworth researched various areas of life and found grit was the key factor for success in all of them.*

## EXPLORE MORE

Watch Duckworth's full TED Talk at TED.com. What did her research in the Chicago's public schools reveal? Share what you learned with your class.

# Project



Students work together in a village school in Burma.

# Project

**A. You are going to find out what factors people think lead to success in life. Work with a partner and follow the steps below.**

1. Brainstorm factors you think might lead to success at school or work.  
For example, grit, I.Q., money, or luck.
2. Ask four people you know what they think are the main factors that lead to success at school or work.
3. Report your findings to the class.

**B. As a class, discuss the most common factors that people thought led to success.**

## EXPLORE MORE

In Lesson A you learned about Rita Pierson. Watch Pierson's TED Talk "Every kid needs a champion" at TED.com. What are Pierson's ideas for improving student learning? Share what you find out with the class.

## Part 1

When I was 27 years old, I left a very demanding job in management consulting for a job that was even more demanding: teaching. I went to teach seventh graders math in the New York City public schools. And like any teacher, I made quizzes and tests. I gave out homework assignments. When the work came back, I calculated grades.

What struck me was that I.Q. was not the only difference between my best and my worst students. Some of my strongest performers did not have stratospheric I.Q. scores. Some of my smartest kids weren't doing so well.

And that got me thinking. The kinds of things you need to learn in seventh grade math, sure, they're hard: ratios, decimals, the area of a parallelogram. But these concepts are not impossible, and I was firmly convinced that every one of my students could learn the material if they worked hard and long enough.

After several more years of teaching, I came to the conclusion that what we need in education is a much better understanding of students and learning from a motivational perspective, from a psychological perspective. In education, the one thing we know how to measure best is I.Q., but what if doing well in school and in life depends on much more than your ability to learn quickly and easily?

## Part 2

So I left the classroom, and I went to graduate school to become a psychologist. I started studying kids and adults in all kinds of super challenging settings, and in every study my question was, who is successful here and why? In all those very different contexts, one characteristic emerged as a significant predictor of success. And it wasn't social intelligence. It wasn't good looks, physical health, and it wasn't I.Q. It was grit.

Grit is passion and perseverance for very long-term goals. Grit is having stamina. Grit is sticking with your future, day in, day out, not just for the week, not just for the month, but for years, and working really hard to make that future a reality. Grit is living life like it's a marathon, not a sprint. . . .

Every day, parents and teachers ask me, “How do I build grit in kids? What do I do to teach kids a solid work ethic? How do I keep them motivated for the long run?” The honest answer is, I don’t know. What I do know is that talent doesn’t make you gritty. Our data show very clearly that there are many talented individuals who simply do not follow through on their commitments. In fact, in our data, grit is usually unrelated or even inversely related to measures of talent.

So far, the best idea I’ve heard about building grit in kids is something called “growth mindset.” This is an idea developed at Stanford University by Carol Dweck, and it is the belief that the ability to learn is not fixed, that it can change with your effort. Dr. Dweck has shown that when kids read and learn about the brain and how it changes and grows in response to challenge, they’re much more likely to persevere when they fail, because they don’t believe that failure is a permanent condition.

So growth mindset is a great idea for building grit. But we need more. And that's where I'm going to end my remarks, because that's where we are. That's the work that stands before us. We need to take our best ideas, our strongest intuitions, and we need to test them. We need to measure whether we've been successful, and we have to be willing to fail, to be wrong, to start over again with lessons learned.

In other words, we need to be gritty about getting our kids grittier.

Thank you.